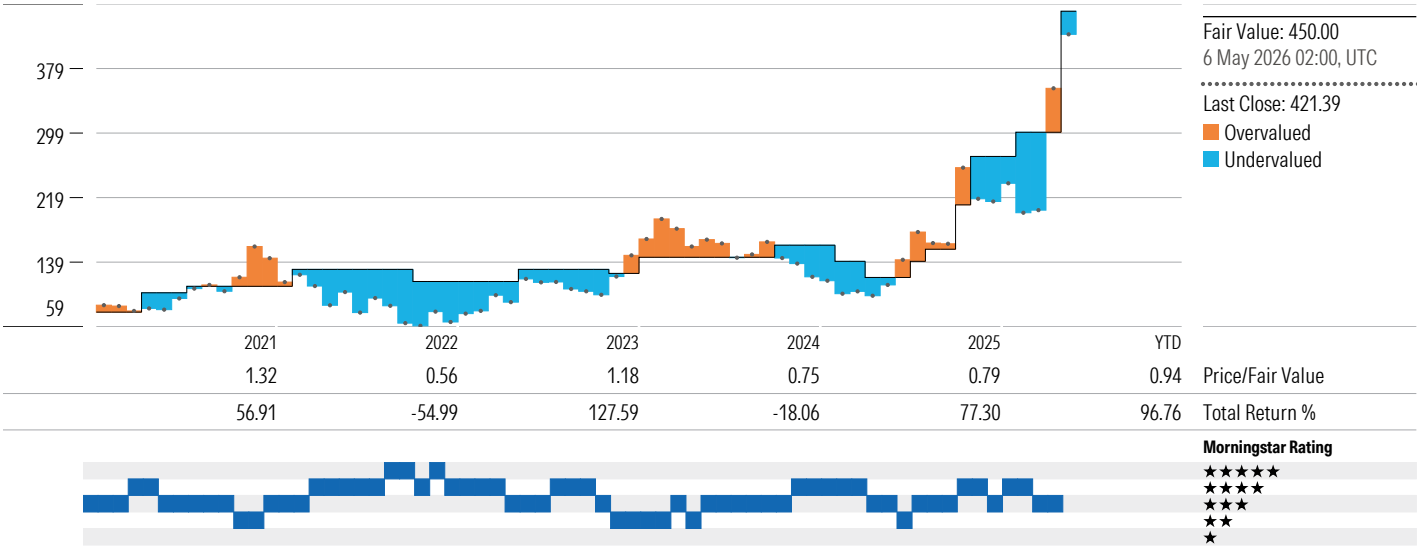
Advanced Micro Devices Inc AMD ★★★ 6 May 2026 23:23, UTC

Last Price	Fair Value Estimate	Price/FVE	Market Cap	Economic Moat™	Equity Style Box	Uncertainty	Capital Allocation	ESG Risk Rating Assessment[1]
421.39 USD 6 May 2026	450.00 USD 6 May 2026 07:00, UTC	0.94	687.01 USD Bil 6 May 2026	Narrow	Large Growth	Very High	Exemplary	5 May 2026 05:00, UTC

Price vs. Fair Value



Total Return % as of 06 May 2026. Last Close as of 06 May 2026. Fair Value as of 6 May 2026 02:00, UTC.

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The primary analyst covering this company does not own its stock.

[1]The ESG Risk Rating Assessment is a representation of Sustainalytics' ESG Risk Rating.

AMD Earnings: Server CPU Demand Is Skyrocketing; Raising Fair Value to $450 From $300

Analyst Note Brian Colello, CPA, Senior Equity Analyst, 6 May 2026

AMD reported first-quarter revenue of $10.25 billion, up 38% year over year and above the high end of guidance. AMD expects June-quarter revenue to be $11.2 billion at the midpoint, up 45% year over year and above the FactSet consensus estimate of $10.5 billion.

Why it matters: AMD reported a blowout quarter thanks to robust server processor (CPU) demand used in agentic AI. The highlight was AMD doubling its forecast for the server CPU market versus its forecast just six months ago. The market should now rise at a 35% compound annual growth rate to $120 billion by 2030.

- AMD disclosed that its server CPU business, which makes up about half of its data center revenue segment, grew more than 50% year over year, with 50%-plus growth to both cloud and enterprise server customers.
- Meanwhile, AMD's closely watched artificial intelligence processor (GPU) market held up well, down slightly on a sequential basis due to lower sales into China. However, we're pleased that AMD remains on pace to bring its important Helios MI450 AI rack-scale solutions to market toward the end of 2026.

The bottom line: We raise our fair value estimate for narrow-moat AMD to $450 from $300. We retain our Very High Morningstar Uncertainty Rating, given the warp speed pace of AI development as agentic

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Sector
Technology

Industry
Semiconductors

Business Description

Advanced Micro Devices designs a variety of digital semiconductors for markets such as PCs, gaming consoles, data centers (including artificial intelligence), industrial, and automotive applications. AMD's traditional strength was in central processing units and graphics processing units used in PCs and data centers. However, AMD is emerging as a prominent player in AI GPUs and related hardware. Additionally, the firm supplies the chips found in prominent game consoles such as the Sony PlayStation and Microsoft Xbox.

AI and server CPUs went from an afterthought to a critical component in short supply in just a few months.

- Shares rose as much as 15% after hours, and we think the rise is justified, given the massive expansion ahead in server CPUs. We've long thought AMD's CPU business and its share gains over Intel were underappreciated, but we underestimated the potential for CPU growth in agentic AI.
- Looking ahead, we think AMD remains well-positioned as a leading x86 server CPU provider. More CPU cores need to be brought to market to orchestrate the exponential rise in AI agents as they connect AI GPU-based servers and traditional data centers.

Business Strategy & Outlook Brian Colello, CPA, Senior Equity Analyst, 6 May 2026

Advanced Micro Devices has significant digital semiconductor expertise and is well positioned to prosper from favorable trends in data centers and artificial intelligence. We consider AMD to be one of two notable firms in graphics processing units, which are especially suited for AI. The company may play second fiddle to Nvidia in AI GPUs, but its GPU expertise should become increasingly valuable—and lucrative—in the years ahead.

We think AMD's data center business should boom over the next few years. Its server CPUs should be in high demand, especially as agentic AI usage rises, and we think it'll carve out a good position in GPUs for generative AI workloads. Although we foresee Nvidia capturing the bulk of the AI market over the next several years, we think that all AI vendors and customers will seek alternatives to keep Nvidia's dominance at bay, and AMD might be the best-positioned to emerge as a second source in AI. AMD's partnerships with OpenAI and Meta should validate AMD's emergence as an AI product leader, in our view.

Historically, AMD's primary products include processors and GPUs tailored to PCs, game consoles, and servers. In our view, AMD's PC and server success stems from the rare x86 architecture license that it possesses from Intel, which allows AMD and Intel to build x86 CPUs for Microsoft Windows PCs. We view it as a heavy lift for Windows to rewrite its x86 software to work with other processors, but Apple made this move in recent years to support its internal ARM-based processors. ARM will likely gain share in the PC market, but we still expect x86-based chips from AMD and Intel to retain leadership in the Windows PC market for quite some time.

AMD has benefited from its outsourced manufacturing model, as its tight relationship with industry leader Taiwan Semiconductor enabled AMD to grab a technological lead as rival Intel stumbled with its internal manufacturing road map. We anticipate that AMD will continue to gain market share over the next few years as Intel strives to turn around, but AMD's gains could last longer if Intel were to stumble further.

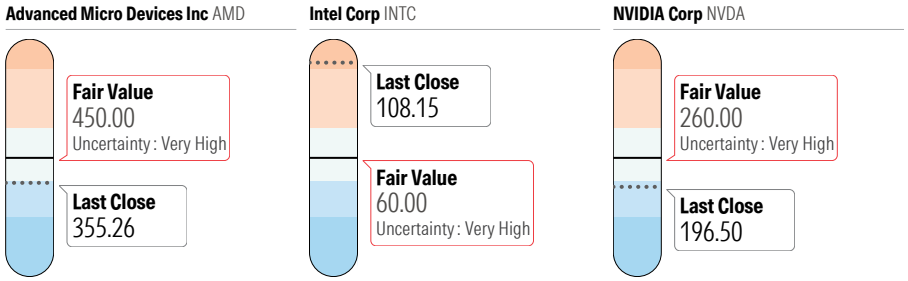
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Competitors



	Advanced Micro Devices Inc AMD	Intel Corp INTC	NVIDIA Corp NVDA
Economic Moat	Narrow	None	Wide
Currency	USD	USD	USD
Fair Value	450.00 5 May 2026 17:43, UTC	60.00 5 May 2026 17:46, UTC	260.00 5 May 2026 17:44, UTC
1-Star Price	787.50	105.00	455.00
5-Star Price	225.00	30.00	130.00
Assessment	Fairly Valued 6 May 2026	Overvalued 6 May 2026	Undervalued 6 May 2026
Morningstar Rating	★★★5 May 2026 05:00, UTC	★5 May 2026 05:00, UTC	★★★★5 May 2026 05:00, UTC
Analyst	Brian Colello, Senior Equity Analyst	Brian Colello, Senior Equity Analyst	Brian Colello, Senior Equity Analyst
Capital Allocation	Exemplary	Standard	Exemplary
Price/Fair Value	1.18	1.80	0.76
Price/Sales	16.13	8.40	22.53
Price/Book	8.84	4.32	30.67
Price/Earning	130.86	—	40.51
Dividend Yield	0.00%	0.00%	—
Market Cap	687.01 Bil	567.99 Bil	5,050.27 Bil
52-Week Range	96.88—430.60	18.97—113.50	110.82—216.83
Investment Style	Large Growth	Large Growth	Large Growth

Bulls Say Brian Colello, CPA, Senior Equity Analyst, 6 May 2026

- AI offers a massive opportunity to GPU makers, and while AMD lags industry leader Nvidia, we see plenty of room in the AI market for AMD.
- AMD has gained share in the PC CPU market as Intel's manufacturing prowess has hit several bumps in recent years.
- AMD's partnership with chip manufacturing leader TSMC, plus its adoption of a chiplet manufacturing strategy, has allowed the company to come to market with more formidable products and greater flexibility to bring new products to market quickly.

Bears Say Brian Colello, CPA, Senior Equity Analyst, 6 May 2026

- AMD will need to improve its software capabilities to make a dent in Nvidia's AI dominance, as Nvidia is strong in not only GPUs but also associated AI software tools.

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- Despite AMD's recent share gains, Intel remains the industry leader in PC CPUs and might recapture most of the market if it can deliver industry-leading manufacturing capabilities once again.
- AMD's gaming semicustom chip business is beholden to the design cycles and launches of new gaming consoles, and it might be a couple of more years until next-generation consoles arrive.

Economic Moat Brian Colello, CPA, Senior Equity Analyst, 6 May 2026

We assign AMD a Narrow Morningstar Economic Moat Rating based on intangible assets around a variety of chip designs (including those from Xilinx, which we rated as having a narrow moat before it was acquired in 2022). We think it is more likely than not that AMD will generate excess returns on capital over the next 10 years, and perhaps even longer.

AMD is perhaps best known for its CPUs for PC desktops and notebooks within its client business segment. We think that AMD has a moat in this business, not only because of chip design expertise honed over decades, but also because it is one of two prominent firms to hold an x86 instruction set architecture license. AMD is also emerging as a prominent GPU vendor in artificial intelligence and holds valuable expertise in hardware, software, and rack-scale solutions, as well.

In PCs, Intel initially developed x86 in the 1970s and licensed the architecture to AMD to satisfy conditions from IBM that required a second source of chips. Intel and AMD are the primary licensees of x86 and owners of x86 intellectual property, as they have a cross-licensing agreement that covers the x86 patent portfolio and would be terminated if either firm were acquired or had a change of control. The exclusivity of the x86 license serves as a significant barrier to entry for other chipmakers (such as Qualcomm and Apple) that cannot make an x86 processor without a license.

For many years, effectively all PC software (including Microsoft's Windows, Apple's Mac operating systems, and any desktop software applications running on both operating systems) was designed for the x86 architecture. Software makers could not easily deviate from building x86-optimized applications, while Microsoft and Apple would have a heavy lift to convert their operating systems to alternate instruction sets (such as those by ARM, which is the dominant instruction set for mobile devices, used in processors from Qualcomm, Samsung, Apple's iPhone, and others). In turn, Apple and PC vendors had little choice but to buy x86 processors from (mostly) Intel or (on occasion) AMD to run their PCs.

Today, however, Apple has already done the heavy lifting of converting its Mac software to ARM and has now built excellent in-house processors for its Mac lineup. Further, an increasing portion of software is hosted in the cloud and can run on multiple operating systems and isn't tied to only x86 devices. Microsoft has dabbled in ARM-based versions of Windows and may increase these investments over time. We still think it would be a heavy lift for any on-device software makers to port their software to be compatible with ARM-based PCs, but we think the writing is on the wall here to move such software to

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the cloud or to ARM-based devices like the Mac. Thus, we don't think the x86 architecture is as moaty as it once was, although we still foresee x86-based processors from Intel and AMD as making up a significant portion of the PC market for the next several years.

In years past, Intel dominated the PC market, not only because of its x86 know-how, but also its internal manufacturing advantage over virtually any other chipmaker, which allowed Intel to come out with best-of-breed products that made it nearly impossible for AMD to gain meaningful market share. These dynamics have flipped in recent years, however, as Intel has stumbled in internal manufacturing. Meanwhile, AMD has partnered closely with Taiwan Semiconductor, the world's largest outsourced chip manufacturer. TSMC now has a market lead in making cutting-edge chips, and AMD is bringing more competitive products to the PC market. Also, AMD has focused on a chiplet strategy with TSMC that is improving yields and flexibility, boosting AMD's portfolio even further.

In turn, AMD has gained market share from Intel in PCs in recent years. Intel is investing heavily in new process nodes to strive for manufacturing parity, if not leadership, with TSMC. Intel also struck a chiplet partnership with Nvidia, where Nvidia will provide GPU tiles within Intel's CPUs. If Intel is successful, it could regain some lost x86 PC market share. However, we no longer view AMD as an also-ran in PC CPUs, and even if Intel were to improve, we still think AMD could retain enough customers to maintain a healthy PC business.

In server CPUs within AMD's data center segment, we see similar dynamics where most of the software (particularly in the enterprise) is based on x86, and we think AMD has a moat in this segment too. Again, Intel's manufacturing stumbles and AMD's partnership with TSMC have enabled AMD to gain market share and profitability, and we don't think Intel will easily recapture such share even if its internal manufacturing were to improve.

We also think that any cloud-based software vendors face high switching costs to move their software from x86 servers to ARM-based servers, although the switching costs are likely far lower between hosting software on an Intel x86 server versus an AMD one. Still, like the PC market, we see ARM-based processors as a threat, as Amazon, Nvidia, and others are developing ARM-based processors to run traditional server workloads, as well as AI applications. Optimizing data center architectures and workloads for ARM-based servers, rather than x86 ones, is again a heavy lift, from what we can gather. Still, we recognize that mega-cap tech firms and cloud leaders (Microsoft, Apple, Amazon) may have the incentive to do so if they can use ARM-based processors to elevate their data centers and devices. Ultimately, we don't foresee a mass exodus away from x86 architectures in the data center for the next several years.

In AI, we think AMD's GPU expertise is becoming increasingly valuable in AI applications. GPUs perform parallel processing, in contrast to the serial processing performed by CPUs, to run the software and

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applications on PCs. In the past decade, the parallel processing of GPUs was found to more efficiently run the matrix multiplication algorithms needed to power artificial intelligence models. AMD is working diligently to deliver AI-centric GPUs to market.

However, Nvidia has a clear lead in GPU hardware and has captured most AI training workloads today. It has spent a decade or more developing its CUDA software platform, creating and hosting a variety of libraries, compilers, frameworks, and development tools that allowed AI professionals to build their models. CUDA is proprietary to Nvidia and only runs on Nvidia GPUs, so we think it will take some effort for AI customers to port any AI models based on CUDA over to AMD's hardware and ROCm software stack.

That said, since Nvidia and CUDA are clearly dominant in AI GPUs today, we think leading cloud vendors will continue to seek second sources for AI and have the necessary incentives to port AI models from Nvidia to other vendors, at least enough to keep Nvidia honest as a vendor. AMD's partnership with OpenAI bodes well on this front. In-house chip development by the hyperscalers will continue, but partnerships with AMD may emerge as well, assuming AMD can improve upon its software chops to allow new customers to use AMD's GPUs for AI training.

We view AMD's embedded business as moaty, as this business is mostly inorganic from the acquisition of Xilinx. We long believed that Xilinx had a narrow moat, based on its expertise in the field-programmable gate array niche of the broader chip industry (with Intel-Altera being the other major FPGA supplier). Although FPGAs are standard components that can be sold to many different users for a multitude of applications, converting a complex algorithm to an efficient custom hardware architecture can be an arduous task. Customers are generally reluctant to switch FPGA vendors, which would require their engineers to learn new software and design tools. Some of Xilinx's FPGAs are used extensively in communication equipment that may have operational lifetimes of at least a decade if not longer. Thus, it can not only be difficult for new entrants to gain market share but also less likely that existing customers defect from Altera (owned by Intel) to Xilinx or vice versa.

We generally do not view AMD's discrete GPU business for gaming nor its semicustom chip business for gaming consoles as moaty, although we acknowledge that the chip design expertise for both likely aided AMD in its ability to deliver profitable GPUs to the data center in the future. In gaming, we think AMD has intangible assets related to GPU patents and chip design expertise. However, it did not lead to excess returns on capital for AMD when its CPU business was lagging, and we believe AMD's GPU operating margins lag its CPU business.

Fair Value and Profit Drivers Brian Colello, CPA, Senior Equity Analyst, 6 May 2026

Our fair value estimate is $450 per share, which implies a 2026 adjusted price/earnings ratio of 70 times but a 2028 and 2030 adjusted price/earnings ratio of 26 times and 15 times, respectively.

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We expect AMD to achieve a top-line compound annual growth rate of 37% from 2026 to 2030, modestly ahead of AMD's forecast of 35% revenue growth over the next three to five years. We attribute our optimism to the rise of agentic AI, which should require exponentially more server CPUs for agentic orchestration workloads.

We model 36% growth in 2026, thanks to the rise of agentic AI driving tremendous demand for server CPUs. We also think AMD will increase its AI GPU business nicely toward the end of 2026 when it brings its Helios rack-scale solutions to market.

We model growth of 54% and 40% in 2027 and 2028, respectively, as partnerships with OpenAI and Meta ramp up and AMD starts to earn tens of billions of dollars in revenue from AI GPUs. Agentic AI should again support strong server CPU growth too. We model longer-term growth rates of 30% in 2029, 26% in 2030, and 9%-11% in the years thereafter as AMD cements its place as a prominent AI vendor.

We are most bullish on the data center segment, in which AMD forecasts a total addressable market of $1 trillion by 2030. This TAM includes GPUs, CPUs, high-bandwidth memory, and AI networking chips.

For AMD's data center segment, which includes both x86 server CPUs and AI GPUs, we model a 53% revenue CAGR over the next five years. Within this segment, we think AMD can carve out a decent portion of the AI accelerator market over time.

We estimate AMD's AI GPU revenue reached $7.4 billion in 2025 versus $5.0 billion in 2024 and nearly zero in 2023 and prior. This $7.4 billion figure is depressed as AMD was banned from selling MI308 products into China in June 2025, and we don't model this revenue reentering our model.

We model AMD reaching $14.5 billion of AI GPU revenue in 2026, boosted by the ramp of its MI450 rack-scale products into OpenAI and Meta in the fourth quarter of 2026. These two partnerships lead us to double our AI GPU revenue estimates thereafter.

We model 107% growth to nearly $30 billion in 2027, 60% growth to $48 billion in 2028, 40% growth to $67 billion in 2029, and 30% growth to $87 billion in 2030. This equates to a 68% CAGR from 2026 to 2030. AMD's forecast calls for an 80% CAGR over the next three to five years in AI GPUs, and we think they'll hit this CAGR in the three-year stretch from 2025 to 2028, but perhaps short over a five-year window.

In client revenue (PC CPUs), we model a 6% CAGR over the next five years as AMD gains market share from Intel, although high memory chip prices may stunt PC production and demand for AMD's PC CPUs. We model an 11% CAGR for AMD's embedded business, albeit off of a lower-than-usual base after a couple of years of revenue declines.

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Adjusted gross margin expanded from 45% in 2020 to 52% in 2025. We model an increase to 58% in 2030, at the high end of management's long-term target of 55%-58%, which was provided before the announcement of the large partnership with Meta. Similarly, adjusted operating margin was 22% in 2025, but data center growth should lead to operating margin expansion to 40% in 2030. This is also above management's long-term target of 35%, although this target was again provided before the Meta deal, which we think will bring operating leverage and improved profitability to AMD.

Risk and Uncertainty Brian Colello, CPA, Senior Equity Analyst, 6 May 2026

Our Morningstar Uncertainty Rating for AMD is Very High, as AI is moving at warp speed, and future workloads and improvements are uncertain. We also see uncertainty around potential tariffs and restrictions on AI sales into China.

AMD sees a massive opportunity to gain share in GPUs targeting AI applications, but we view Nvidia as a clear leader here with a wide moat in not only hardware design but also associated software tools. Even if AMD's GPU designs are up to par (or better), we view the associated software tools as a hurdle where AMD is behind today and will need to catch up to Nvidia. Further, we expect leading hyperscale cloud computing customers to continue to invest in AI processors. Google's Tensor processing units and Amazon's Trainium and Inferentia chips were designed with AI workloads in mind, while Microsoft and Meta have announced semiconductor design plans.

In PCs, AMD continues to square off against Intel, the formerly dominant market leader that has lost its manufacturing edge in recent years. Intel struck a partnership with Nvidia, which might make for improved PC products. If Intel can regain its manufacturing lead, AMD may face a more formidable x86 foe. Additionally, if Microsoft were to continue to push its Windows PC ecosystem toward PCs with ARM-based processors, new entrants could pose credible threats to both AMD and Intel. All the while, the PC market remains cyclical, and AMD will have to navigate the cycles accordingly.

On the environmental, social, and governance front, we do not foresee any material issues on the horizon. Perhaps the greatest risk is the potential scarcity of experienced chip design talent. We think the firm has done a solid job developing chip designers internally.

Capital Allocation Brian Colello, CPA, Senior Equity Analyst, 6 May 2026

We assign AMD an Exemplary Morningstar Capital Allocation Rating. The rating reflects our assessments of a sound balance sheet, exceptional investments associated with the firm's strategy and execution, and attractive and appropriate shareholder distribution policies.

Lisa Su took over as CEO in 2014; she was previously COO. She was named chair of the board as well in 2022. In 2023, Jean Hu became CFO, having served as Marvell's CFO previously. Since Su took charge, we think management has done a solid job in driving a more focused long-term plan across both PC and server end markets. The firm has adopted a chiplet strategy (which disaggregates parts of chips

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MORNINGSTAR®

Advanced Micro Devices Inc AMD ★★★ 6 May 2026 23:23, UTC

Last Price	Fair Value Estimate	Price/FVE	Market Cap	Economic Moat™	Equity Style Box	Uncertainty	Capital Allocation	ESG Risk Rating Assessment[1]
421.39 USD 6 May 2026	450.00 USD 6 May 2026 07:00, UTC	0.94	687.01 USD Bil 6 May 2026	Narrow	Large Growth	Very High	Exemplary	5 May 2026 05:00, UTC

and utilizes the most practical manufacturing process) and tapped TSMC to manufacture its leading-edge processors. As a result of this strategy, AMD has been capturing market share at Intel's expense as the latter has struggled with its advanced manufacturing processes. We are also impressed by the pace of AMD's innovation in AI.

We think AMD has made nice strategic moves in recent years to expand beyond the PC CPU market. In February 2022, AMD closed the purchase of FPGA leader Xilinx in an all-stock deal worth about $50 billion. Xilinx represents a margin-accretive business and gives AMD entry into industrial, automotive, and communications infrastructure end markets, all while helping the company offer more robust products in the data center. AMD has also made nice bolt-on acquisitions, such as Pensando Systems, Silo AI, and ZT Systems, to improve its presence in AI and data centers. Organically, we like AMD's focus on the data center end market, as the company is well positioned to carve out a piece of the AI pie with its GPU expertise.

We view AMD as having a solid balance sheet. As of March 2026, AMD had $12.3 billion in cash and cash equivalents against total debt of $3.2 billion. AMD does not pay a dividend but has bought back shares in recent years. We'd expect any capital distributions in the years ahead to be done via additional buybacks as part of this program.

Analyst Notes Archive

AMD: Raising Fair Value to $300 From $270 as Meta Agrees to Be AMD's Second Large AI Customer

Brian Colello, CPA,Senior Equity Analyst,24 Feb 2026

Advanced Micro Devices and Meta Platforms announced an expanded strategic partnership in which Meta may deploy up to 6 gigawatts of AMD customized GPU solutions, with the first gigawatt expected to begin in the second half of 2026. Meta has the option to acquire 160 million AMD shares via warrants.Why it matters: AMD's deal with Meta resembles its flagship deal with OpenAI, both in gigawatt size and warrant issuance/timing. AMD has long hinted that it has other customers beyond OpenAI, and we view this deal as being largely in line with our expectations for its expansion. The bottom line: We raise our fair value estimate for narrow-moat AMD to $300 per share from $270 as we boost our AI graphics processing unit revenue assumptions, partially offset by the potential dilution from the warrants. The shares rose nearly 9% on Feb. 24 but still appear undervalued. We suspect our disconnect with market prices stems from pessimism about OpenAI's expansion. AMD's deal with Meta helps to derisk AMD from being highly exposed to OpenAI specifically or a single model builder generally. That said, our revenue estimates do not fully incorporate 6 GW from both OpenAI and Meta by 2030, so we can envision further upside if both model builders deploy artificial intelligence as aggressively and as quickly as expressed in their press releases.Coming up: AMD doesn't expect much revenue from the Meta deal in 2026, as sales should start in the fourth quarter. Revenue should accelerate from there. The deal brings Meta closer to reaching its previous guidance for AI growth at an

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80% compound annual rate over the next three to five years. The customized product for Meta appears to be at the chip and system level but doesn't appear to be revolutionary. AMD thinks the AI revenue will be at similar gross margin levels to the rest of its AI business, while additional operating costs to serve Meta are minimal.

AMD Earnings: Data Center Demand Still Looks Quite Strong to Us; Shares Undervalued Brian Colello, CPA,Senior Equity Analyst,4 Feb 2026

Advanced Micro Devices reported fourth-quarter revenue of $10.3 billion, up 34% year over year, up 11% sequentially, and ahead of guidance. AMD's first-quarter revenue forecast of $9.8 billion is ahead of FactSet consensus estimates and would represent 32% growth year over year. Why it matters: AMD received a revenue boost of $390 million from previously-banned MI308 chip sales into China. Still, AMD's core data center chip business fared well with sequential growth in artificial intelligence accelerators (GPUs) and strong demand for x86 server processors (CPUs) for traditional workloads. AMD reiterated its long-term financial targets, first provided at its November 2025 analyst day, calling for a 35% revenue compound annual growth rate over the next three to five years. Within, AMD still expects a 60% CAGR in data center revenue, aiming for $100 billion by 2030 versus $16.6 billion in 2025. The bottom line: We maintain our $270 fair value estimate for narrow-moat AMD. Shares sold off about 7% after hours, and like the selloff in Intel's shares after earnings, our best guess is that investors may have been too optimistic about the revenue growth seen in server CPUs. Medium-term concerns about OpenAI's expansion are still likely an overhang on AMD's stock, but it appears that the hard work is being done to bring AMD's MI450 products to OpenAI this year, starting in the third quarter, with meaningful revenue arriving in the fourth quarter. Coming up: By end market, on a sequential basis, AMD expects first-quarter revenue growth in data center but declines across client (PCs), gaming, and embedded, the latter three of which typically face seasonal first-quarter declines. AMD hinted that server CPU revenue will grow sequentially, which is stronger than normal seasonal patterns. GPU revenue should also rise sequentially, but we suspect this growth will be moderate, given the one-time boost in the previous quarter from MI308 revenue sold into China.

AMD: Maintain $270 Fair Value Estimate Amid Reports of Meta's Negotiations to Buy Google TPUs Brian Colello, CPA,Senior Equity Analyst,25 Nov 2025

The Information reported that Meta Platforms is in discussions to buy Google's custom Tensor Processing Unit, or TPU, chips for artificial intelligence. Shares of AI chip suppliers Nvidia and AMD were down as much as 4% and 7%, respectively, on Nov. 25.Why it matters: Google's TPUs appear to be gaining traction outside of Google's internal use cases. We've long expected hyperscalers like Meta to develop custom AI processors (ASICs), but it's possible that Meta could diminish its chip development and turn to Google for ASICs instead. Meta is a large customer of Nvidia and an important emerging partner with AMD, the latter of which also aspires to gain AI processor share. These reports raise the

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risk, but not the certainty, that TPUs displace AI chips from these other two suppliers. However, we've long believed in an AI world where Nvidia, AMD, and ASICs all have a place. These discussions don't change this dynamic, in our view. That said, the relative market share of chips from all three sources is uncertain and will be based on future innovation and execution. The bottom line: We maintain our fair value estimates of $240 for wide-moat Nvidia and $270 for narrow-moat AMD. We retain our Very High Uncertainty Ratings for both firms because AI dynamics are changing rapidly and based on opaque negotiations among tech leaders. For Nvidia, we expect it to inevitably lose market share to ASICs like TPUs in certain use cases. However, the firm's dominance in AI software and networking around these chips should keep Nvidia as the AI vendor of choice and the industry leader. The reports are perhaps more worrisome for AMD, which needs to prove its AI capabilities to hyperscalers. That said, we view these reports are more likely a shift in Meta's ASIC strategy to TPUs, rather than a signal that the Meta-AMD partnership is doomed.

AMD: Investor Day Touts Tremendous AI Growth With Steady Margins; Fair Value Up to $270 from $210 Brian Colello, CPA,Senior Equity Analyst,12 Nov 2025

AMD hosted an investor day with updated revenue growth and financial targets. These include growth in the next three to five years of 80% in data center AI products, 60% in all data center products, and 35% for total AMD. AMD's adjusted gross margin target is now 55%-58%, ahead of 54% currently. Why it matters: AMD increased its bullishness on the artificial intelligence market, both in total (targeting a $1 trillion-plus market by 2030) and for the company (expecting 10%-plus market share). This implies $100 billion in AI revenue for AMD over the next three to five years, which exceeds our expectations. Meanwhile, AMD is targeting modest gross margin expansion, which is encouraging to us as we were concerned that AI revenue might be a bit dilutive as the company strives to gain market share. Between growth and healthy margins, AMD is targeting $20 of earnings per share by 2030. AI growth seems plausible to us and is based on its customer conversations. AMD has public partnerships with OpenAI, Oracle, and Meta today. AMD hinted that it is in deep discussions with other leading hyperscalers, sovereign entities, and AI-native firms. The bottom line: We raise our fair value estimate for narrow-moat AMD to $270 from $210, as we once again lift our AI revenue estimates for the firm. We retain our Very High Uncertainty Rating as the AI market continues to shift rapidly. Shares now appear a little undervalued to us. We now model a 31% revenue compound annual growth rate for AMD through 2029, up from our prior estimate of 26%. Our data center and AI GPU CAGRs now rise to 42% and 62%, respectively, up from 37% and 55%. Yet we still have some modest conservatism compared with AMD's targets. Even though the AI industry is computing-constrained, and management addressed these concerns head-on, we're still cautious that industry funding and energy/power generation might cause firms like OpenAI to grow a bit more slowly than visualized.

AMD Earnings: Strong, but We Think Investors Wanted Greater AI Optimism; $210 Fair Value Kept

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Brian Colello, CPA,Senior Equity Analyst,5 Nov 2025

AMD reported strong third-quarter revenue, up 20% sequentially, up 36% year over year, and above the high end of guidance. AMD's fourth-quarter revenue forecast of $9.6 billion would represent growth of 4% sequentially and 25% year over year, and was ahead of FactSet consensus estimates.Why it matters: AMD continues to execute well with strong growth in client (PC) revenue, server processors (CPUs), and artificial intelligence products. Meanwhile, AMD is optimistic that its OpenAI deal will generate "well over $100 billion" in revenue in the next few years. Data center revenue of $4.3 billion rose 34% sequentially and 22% year over year, thanks to a healthy CPU server market and share gains over Intel. AMD's AI business also fared well, and our initial guess is that both servers and AI products grew a little over 20% year over year. Client revenue from the PC end market grew 10% sequentially and 46% year over year, as AMD is again gaining share over Intel across desktops, gaming PCs, and PCs sold into enterprises. The bottom line: We maintain our $210 fair value estimate for narrow-moat AMD and retain our Very High Morningstar Uncertainty Rating as vastly different scenarios may play out for AMD's AI business over time. Shares fell about 3% after hours, and we suspect investors were seeking both greater insight and greater optimism into AMD's AI business. Shares still appear a bit overvalued to us. We think AMD's comments around $100 billion-plus of OpenAI revenue and "tens of billions" of AI revenue coming in 2027 might be conservative, but have already been baked in by the market at this point. Coming up: AMD expects fourth-quarter data center revenue to rise over 10% sequentially. AMD will host an analyst day next week where we hope to learn more about its AI prospects. We may hear more optimism at this event, and AMD hinted that its prior forecast of $500 billion of industrywide AI processor revenue by 2028 will be revised upward next week.

AMD: The Company Has Arrived in AI With a Massive OpenAI Deal; Fair Value Estimate to $210 From $155 Brian Colello, CPA,Senior Equity Analyst,6 Oct 2025

AMD and OpenAI reached an agreement where AMD will supply artificial intelligence gear for up to 6 gigawatts of OpenAI's AI infrastructure. AMD expects the deal to earn tens of billions of dollars of annual incremental revenue with the rollout starting in second-half 2026. Why it matters: We're highly encouraged with this deal for AMD as it validates its AI technology and significantly boosts its future AI revenue and profits. AMD hinted at the potential for "well over $100 billion" in future revenue. We've long believed that AMD will carve out a piece of the AI pie. This deal suggests that AMD's slice of the market might be a bit higher than expected. However, we doubt it displaces Nvidia as the dominant AI vendor. The bottom line: We raise our fair value estimate for narrow-moat AMD to $210 from $155 thanks to this upcoming AI windfall. We retain our Very High Uncertainty Rating as surprising, massive AI deals seem to be arising weekly. We have doubled our estimates for AMD's AI GPU revenue in 2027 and beyond. We now project $42.2 billion for AMD by 2029 versus our prior estimate of $20.7 billion. AMD will issue warrants for up to 160 million shares (about 10% of total shares) to OpenAI. We're not

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alarmed by the warrants and suspect that investors would gladly trade this accretive revenue windfall for some stock dilution if AMD were to hit all its targets. Coming up: OpenAI continues to have massive aspirations for AI adoption, driving significant deals with both Nvidia and AMD. No timeline was set for OpenAI's rollout. Perhaps the greatest risk to AMD's future revenue from this deal may come from the energy and power requirements that OpenAI and its partners will need for such massive infrastructure buildouts. If OpenAI and its partners were to roll out this infrastructure slower than envisioned, it might be bad for their aspirations but perhaps good for investors fearful of an AI bubble. The 90's .com bubble had no such outside energy constraints.

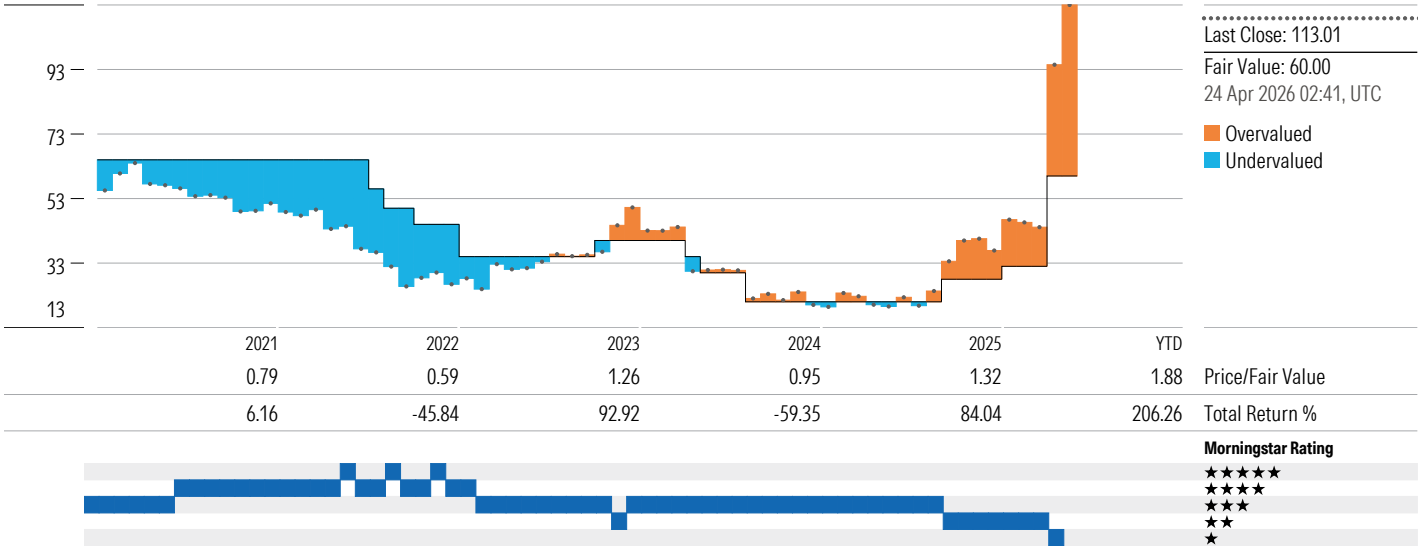
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Competitors Price vs. Fair Value

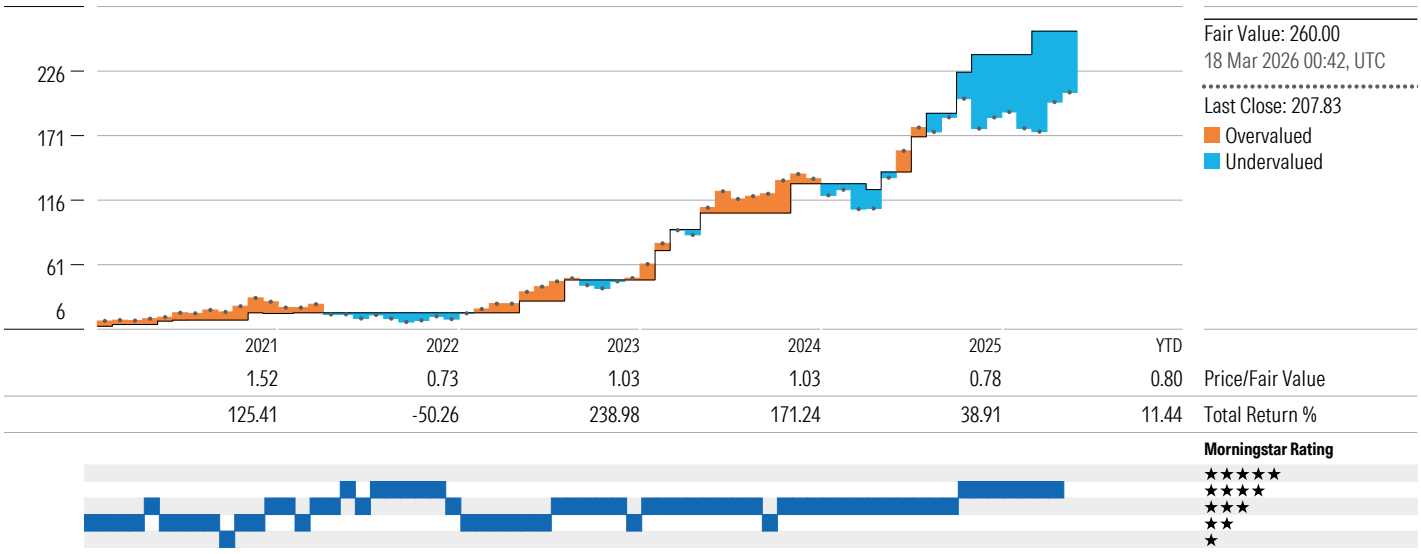
Intel Corp INTC



	2021	2022	2023	2024	2025	YTD	
	0.79	0.59	1.26	0.95	1.32	1.88	Price/Fair Value
	6.16	-45.84	92.92	-59.35	84.04	206.26	Total Return %

Total Return % as of 06 May 2026. Last Close as of 06 May 2026. Fair Value as of 24 Apr 2026 02:41, UTC.

NVIDIA Corp NVDA



	2021	2022	2023	2024	2025	YTD	
	1.52	0.73	1.03	1.03	0.78	0.80	Price/Fair Value
	125.41	-50.26	238.98	171.24	38.91	11.44	Total Return %

Total Return % as of 06 May 2026. Last Close as of 06 May 2026. Fair Value as of 18 Mar 2026 00:42, UTC.

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Morningstar Valuation Model Summary

Financials as of 05 May 2026

Fiscal Year, ends 31 Dec	Actual 2023	2024	2025	Forecast 2026	2027	2028	2029	2030
Revenue (USD Mil)	22,680	25,785	34,639	46,983	72,435	101,070	131,492	165,411
Operating Income (USD Mil)	367	2,038	3,694	7,643	18,220	31,548	45,956	61,984
EBITDA (USD Mil)	3,854	4,964	6,698	10,604	21,163	34,483	48,765	64,893
Adjusted EBITDA (USD Mil)	3,854	4,964	6,698	10,604	21,163	34,483	48,765	64,893
Net Income (USD Mil)	854	1,641	4,335	6,905	16,008	27,604	40,138	54,082
Adjusted Net Income (USD Mil)	4,302	5,420	6,829	10,699	19,809	31,496	44,039	58,208
Free Cash Flow To The Firm (USD Mil)	415	-498	1,172	9,202	12,175	20,813	32,605	45,539
Weighted Average Diluted Shares Outstanding (Mil)	1,625	1,637	1,636	1,658	1,715	1,825	1,935	1,990
Earnings Per Share (Diluted) (USD)	0.53	1.00	2.65	4.17	9.33	15.13	20.74	27.18
Adjusted Earnings Per Share (Diluted) (USD)	2.65	3.31	4.17	6.45	11.55	17.26	22.76	29.25
Dividends Per Share (USD)	0.00	0.00	0.00	0.00	0.00	0.00	0.00	0.00

Margins & Returns as of 05 May 2026

	3 Year Avg	Actual 2023	2024	2025	Forecast 2026	2027	2028	2029	2030	5 Year Avg
Operating Margin %	6.6	1.6	7.9	10.7	16.3	25.2	31.2	34.9	37.5	29.0
EBITDA Margin %	—	17.0	19.3	19.3	22.6	29.2	34.1	37.1	39.2	—
Adjusted EBITDA Margin %	—	17.0	19.3	19.3	22.6	29.2	34.1	37.1	39.2	32.4
Net Margin %	7.6	3.8	6.4	12.5	14.7	22.1	27.3	30.5	32.7	25.5
Adjusted Net Margin %	19.9	19.0	21.0	19.7	22.8	27.4	31.2	33.5	35.2	30.0
Free Cash Flow To The Firm Margin %	1.1	1.8	-1.9	3.4	19.6	16.8	20.6	24.8	27.5	21.9

Growth & Ratios as of 05 May 2026

	3 Year CAGR	Actual 2023	2024	2025	Forecast 2026	2027	2028	2029	2030	5 Year CAGR
Revenue Growth %	13.6	-3.9	13.7	34.3	35.6	54.2	39.5	30.1	25.8	36.7
Operating Income Growth %	47.0	-68.4	455.3	81.3	106.9	138.4	73.2	45.7	34.9	75.8
EBITDA Growth %	11.5	-29.1	28.8	34.9	58.3	99.6	62.9	41.4	33.1	59.1
Adjusted EBITDA Growth %	7.2	-29.1	28.8	34.9	58.3	99.6	62.9	41.4	33.1	57.5
Earnings Per Share Growth %	46.7	-37.5	90.8	164.3	57.2	124.1	62.0	37.1	31.0	59.3
Adjusted Earnings Per Share Growth %	46.7	-24.4	25.1	26.1	54.6	78.9	49.4	31.9	28.5	59.3

Valuation as of 05 May 2026

	Actual 2023	2024	2025	Forecast 2026	2027	2028	2029	2030
Price/Earning	55.6	36.5	51.4	65.3	36.5	24.4	18.5	14.4
Price/Sales	10.5	7.6	10.1	14.6	9.5	6.8	5.2	4.2
Price/Book	4.3	3.4	5.6	9.2	6.5	4.4	3.2	2.6
Price/Cash Flow	—	—	—	—	—	—	—	—
EV/EBITDA	61.0	39.0	51.6	64.0	32.1	19.7	13.9	10.5
EV/EBIT	640.9	95.0	93.6	88.8	37.2	21.5	14.8	10.9
Dividend Yield %	—	—	—	—	—	—	—	—
Dividend Payout %	0.0	0.0	0.0	0.0	0.0	0.0	0.0	0.0
Free Cash Flow Yield %	—	—	—	—	—	—	—	—

Operating Performance / Profitability as of 05 May 2026

Fiscal Year, ends 31 Dec	Actual 2023	2024	2025	Forecast 2026	2027	2028	2029	2030
ROA %	1.3	2.4	5.6	7.5	11.9	13.4	13.8	14.5
ROE %	1.5	2.9	6.9	9.1	14.5	15.8	16.0	16.7
ROIC %	4.7	6.1	8.2	12.8	20.2	24.2	25.2	26.1

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MORNINGSTAR®

Advanced Micro Devices Inc AMD ★★★ 6 May 2026 23:23, UTC

Last Price	Fair Value Estimate	Price/FVE	Market Cap	Economic Moat™	Equity Style Box	Uncertainty	Capital Allocation	ESG Risk Rating Assessment[1]
421.39 USD 6 May 2026	450.00 USD 6 May 2026 07:00, UTC	0.94	687.01 USD Bil 6 May 2026	Narrow	Large Growth	Very High	Exemplary	5 May 2026 05:00, UTC

Financial Leverage (Reporting Currency)	Actual			Forecast				
Fiscal Year, ends 31 Dec	**2023**	**2024**	**2025**	**2026**	**2027**	**2028**	**2029**	**2030**
Debt/Capital %	1.3	1.1	1.1	0.2	0.2	0.2	0.1	0.1
Assets/Equity	1.2	1.2	1.2	1.2	1.2	1.2	1.2	1.2
Net Debt/EBITDA	-0.7	-0.6	-1.0	-2.0	-2.5	-3.2	-3.7	-3.8
Total Debt/EBITDA	0.8	0.4	0.6	0.2	0.1	0.0	0.0	0.0
EBITDA/ Net Interest Expense	-42.4	-55.8	-15.0	-34.4	-117.6	-191.6	-270.9	-360.5

Forecast Revisions as of 5 May 2026	2026		2027		2028	
Prior data as of 24 Feb 2026	**Current**	**Prior**	**Current**	**Prior**	**Current**	**Prior**
Fair Value Estimate Change (Trading Currency)	450.00	300.00	—	—	—	—
Revenue (USD Mil)	46,983	44,734	72,435	64,693	101,070	86,005
Operating Income (USD Mil)	7,643	6,087	18,220	14,965	31,548	25,187
EBITDA (USD Mil)	10,604	9,048	21,163	17,908	34,483	28,122
Net Income (USD Mil)	10,699	9,159	19,809	16,885	31,496	25,756
Earnings Per Share (Diluted) (USD)	4.17	3.31	9.33	7.79	15.13	12.25
Adjusted Earnings Per Share (Diluted) (USD)	6.45	5.60	11.55	9.99	17.26	14.30
Dividends Per Share (USD)	0.00	0.00	0.00	0.00	0.00	0.00

Key Valuation Drivers as of 05 May 2026

Cost of Equity %	9.2
Pre-Tax Cost of Debt %	5.8
Weighted Average Cost of Capital %	9.2
Long-Run Tax Rate %	—
Stage II EBI Growth Rate %	9.0
Stage II Investment Rate %	40.9
Perpetuity Year	15

Additional estimates and scenarios available for download at https://pitchbook.com/.

Discounted Cash Flow Valuation as of 05 May 2026

	USD Mil
Present Value Stage I	263,868
Present Value Stage II	110,170
Present Value Stage III	442,361
Total Firm Value	**816,399**
Cash and Equivalents	10,552
Debt	3,222
Other Adjustments	-100,000
Equity Value	**723,729**
Projected Diluted Shares	1,658
Fair Value per Share (USD)	**450.00**

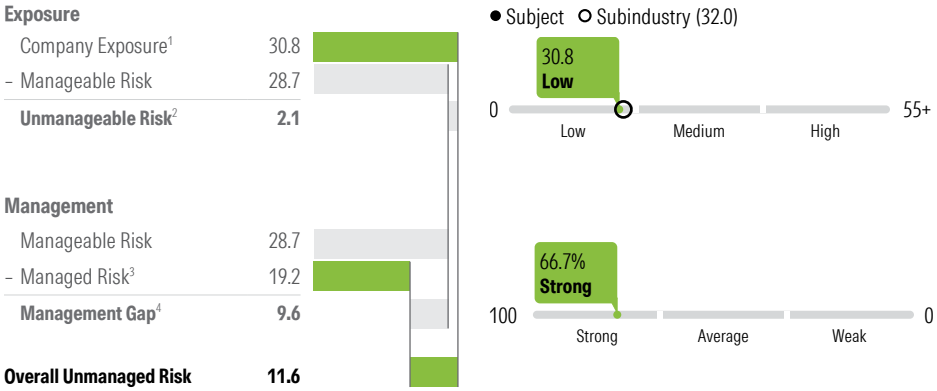
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Advanced Micro Devices Inc AMD ★★★ 6 May 2026 23:23, UTC

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ESG Risk Rating Breakdown



- Exposure represents a company's vulnerability to ESG risks driven by their business model
- Exposure is assessed at the Subindustry level and then specified at the company level
- Scoring ranges from 0-55+ with categories of low, medium, and high-risk exposure

- Management measures a company's ability to manage ESG risks through its commitments and actions
- Management assesses a company's efficiency on ESG programs, practices, and policies
- Management score ranges from 0-100% showing how much manageable risk a company is managing

ESG Risk Rating



ESG Risk Ratings measure the degree to which a company's value is impacted by environmental, social, and governance risks, by evaluating the company's ability to manage the ESG risks it faces.

1. A company's Exposure to material ESG issues 2. Unmanageable Risk refers to risks that are inherent to a particular business model that cannot be managed by programs or initiatives 3. Managed Risk = Manageable Risk multiplied by a Management score of 66.7% 4. Management Gap assesses risks that are not managed, but are considered manageable 5. ESG Risk Rating Assessment = Overall Unmanaged Risk = Management Gap plus Unmanageable Risk

ESG Risk Rating Assessment[5]



ESG Risk Rating is of May 05, 2026. Highest Controversy Level is as of Apr 08, 2026. Sustainalytics Subindustry: Semiconductor Design and Manufacturing. Sustainalytics provides Morningstar with company ESG ratings and metrics on a monthly basis and as such, the ratings in Morningstar may not necessarily reflect current Sustainalytics' scores for the company. For the most up to date rating and more information, please visit: sustainalytics.com/esg-ratings/.

Peer Analysis 05 May 2026

Peers are selected from the company's Sustainalytics-defined Subindustry and are displayed based on the closest market cap values

Company Name	Exposure			Management			ESG Risk Rating		
Advanced Micro Devices Inc	30.8 | Low	0	55+	66.7 | Strong	100	0	11.6 | Low	0	40+
Intel Corp	50.1 | Medium	0	55+	68.3 | Strong	100	0	18.6 | Low	0	40+
NVIDIA Corp	30.2 | Low	0	55+	62.9 | Strong	100	0	12.4 | Low	0	40+
Maxscend Microelectronics Co Ltd	44.8 | Medium	0	55+	25.3 | Average	100	0	34.3 | High	0	40+
—	— | —	0	55+	— | —	100	0	— | —	0	40+

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Appendix

Historical Morningstar Rating

Advanced Micro Devices Inc AMD 5 May 2026 05:00, UTC

Dec 2026	Nov 2026	Oct 2026	Sep 2026	Aug 2026	Jul 2026	Jun 2026	May 2026	Apr 2026	Mar 2026	Feb 2026	Jan 2026
—	—	—	—	—	—	—	★★★	★★★	★★★★	★★★★	★★★
Dec 2025	Nov 2025	Oct 2025	Sep 2025	Aug 2025	Jul 2025	Jun 2025	May 2025	Apr 2025	Mar 2025	Feb 2025	Jan 2025
★★★★	★★★★	★★★	★★★	★★★	★★	★★★	★★★	★★★★	★★★★	★★★★	★★★★
Dec 2024	Nov 2024	Oct 2024	Sep 2024	Aug 2024	Jul 2024	Jun 2024	May 2024	Apr 2024	Mar 2024	Feb 2024	Jan 2024
★★★★	★★★	★★★	★★★	★★★	★★★	★★★	★★	★★★	★★	★★	★★
Dec 2023	Nov 2023	Oct 2023	Sep 2023	Aug 2023	Jul 2023	Jun 2023	May 2023	Apr 2023	Mar 2023	Feb 2023	Jan 2023
★★	★★★	★★★★	★★★★	★★★★	★★★	★★★	★★★	★★★★	★★★★	★★★★	★★★★
Dec 2022	Nov 2022	Oct 2022	Sep 2022	Aug 2022	Jul 2022	Jun 2022	May 2022	Apr 2022	Mar 2022	Feb 2022	Jan 2022
★★★★★	★★★★	★★★★★	★★★★★	★★★★	★★★★	★★★★	★★★★	★★★★	★★★	★★★	★★★
Dec 2021	Nov 2021	Oct 2021	Sep 2021	Aug 2021	Jul 2021	Jun 2021	May 2021	Apr 2021	Mar 2021	Feb 2021	Jan 2021
★★	★★	★★★	★★★	★★★	★★★	★★★	★★★★	★★★★	★★★	★★★	★★★

Intel Corp INTC 5 May 2026 05:00, UTC

Dec 2026	Nov 2026	Oct 2026	Sep 2026	Aug 2026	Jul 2026	Jun 2026	May 2026	Apr 2026	Mar 2026	Feb 2026	Jan 2026
—	—	—	—	—	—	—	★	★★	★★	★★	★★
Dec 2025	Nov 2025	Oct 2025	Sep 2025	Aug 2025	Jul 2025	Jun 2025	May 2025	Apr 2025	Mar 2025	Feb 2025	Jan 2025
★★	★★	★★	★★★	★★★	★★★	★★★	★★★	★★★	★★★	★★★	★★★
Dec 2024	Nov 2024	Oct 2024	Sep 2024	Aug 2024	Jul 2024	Jun 2024	May 2024	Apr 2024	Mar 2024	Feb 2024	Jan 2024
★★★	★★★	★★★	★★★	★★★	★★★	★★★	★★★	★★★	★★★	★★★	★★★
Dec 2023	Nov 2023	Oct 2023	Sep 2023	Aug 2023	Jul 2023	Jun 2023	May 2023	Apr 2023	Mar 2023	Feb 2023	Jan 2023
★★	★★★	★★★	★★★	★★★	★★★	★★★	★★★	★★★	★★★	★★★★	★★★★
Dec 2022	Nov 2022	Oct 2022	Sep 2022	Aug 2022	Jul 2022	Jun 2022	May 2022	Apr 2022	Mar 2022	Feb 2022	Jan 2022
★★★★★	★★★★	★★★★	★★★★★	★★★★	★★★★	★★★★★	★★★★	★★★★	★★★★	★★★★	★★★★
Dec 2021	Nov 2021	Oct 2021	Sep 2021	Aug 2021	Jul 2021	Jun 2021	May 2021	Apr 2021	Mar 2021	Feb 2021	Jan 2021
★★★★	★★★★	★★★★	★★★★	★★★★	★★★★	★★★	★★★	★★★	★★★	★★★	★★★

NVIDIA Corp NVDA 5 May 2026 05:00, UTC

Dec 2026	Nov 2026	Oct 2026	Sep 2026	Aug 2026	Jul 2026	Jun 2026	May 2026	Apr 2026	Mar 2026	Feb 2026	Jan 2026
—	—	—	—	—	—	—	★★★★	★★★★	★★★★	★★★★	★★★★
Dec 2025	Nov 2025	Oct 2025	Sep 2025	Aug 2025	Jul 2025	Jun 2025	May 2025	Apr 2025	Mar 2025	Feb 2025	Jan 2025
★★★★	★★★★	★★★	★★★	★★★	★★★	★★★	★★★	★★★	★★★	★★★	★★★
Dec 2024	Nov 2024	Oct 2024	Sep 2024	Aug 2024	Jul 2024	Jun 2024	May 2024	Apr 2024	Mar 2024	Feb 2024	Jan 2024
★★★	★★★	★★	★★★	★★★	★★★	★★★	★★★	★★★	★★★	★★★	★★
Dec 2023	Nov 2023	Oct 2023	Sep 2023	Aug 2023	Jul 2023	Jun 2023	May 2023	Apr 2023	Mar 2023	Feb 2023	Jan 2023
★★★	★★★	★★★	★★★	★★★	★★	★★	★★	★★	★★	★★	★★★
Dec 2022	Nov 2022	Oct 2022	Sep 2022	Aug 2022	Jul 2022	Jun 2022	May 2022	Apr 2022	Mar 2022	Feb 2022	Jan 2022
★★★★	★★★★	★★★★	★★★★	★★★★	★★★	★★★★	★★★	★★★	★★	★★★	★★★
Dec 2021	Nov 2021	Oct 2021	Sep 2021	Aug 2021	Jul 2021	Jun 2021	May 2021	Apr 2021	Mar 2021	Feb 2021	Jan 2021
★★	★★	★	★★	★★	★★	★★	★★★	★★	★★	★★	★★

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Research Methodology for Valuing Companies

Overview

At the heart of our valuation system is a detailed projection of a company's future cash flows, resulting from our analysts' research. Analysts create custom industry and company assumptions to feed income statement, balance sheet, and capital investment assumptions into our globally standardized, proprietary discounted cash flow, or DCF, modeling templates. We use scenario analysis, indepth competitive advantage analysis, and a variety of other analytical tools to augment this process. Moreover, we think analyzing valuation through discounted cash flows presents a better lens for viewing cyclical companies, high-growth firms, businesses with finite lives (e.g., mines), or companies expected to generate negative earnings over the next few years. That said, we don't dismiss multiples altogether but rather use them as supporting cross-checks for our DCF-based fair value estimates. We also acknowledge that DCF models offer their own challenges (including a potential proliferation of estimated inputs and the possibility that the method may miss shortterm market-price movements), but we believe these negatives are mitigated by deep analysis and our longterm approach.

Morningstar's equity research group ("we," "our") believes that a company's intrinsic worth results from the future cash flows it can generate. The Morningstar Rating for stocks identifies stocks trading at a discount or premium to their intrinsic worth—or fair value estimate, in Morningstar terminology. Five-star stocks sell for the biggest risk adjusted discount to their fair values, whereas 1-star stocks trade at premiums to their intrinsic worth.

Four key components drive the Morningstar rating: (1) our assessment of the firm's economic moat, (2) our estimate of the stock's fair value, (3) our uncertainty around that fair value estimate and (4) the current market price. This process ultimately culminates in our singlepoint star rating.

1. Economic Moat

The concept of an economic moat plays a vital role not only in our qualitative assessment of a firm's long-term investment potential, but also in the actual calculation of our fair value estimates. An economic moat is a structural feature that allows a firm to sustain excess profits over a long period of time. We define economic profits as returns on invested capital (or ROIC) over and above our estimate of a firm's cost of capital, or weighted average cost of capital (or WACC). Without a moat, profits are more susceptible to competition. We have identified five sources of economic moats: intangible assets, switching costs, network effect, cost advantage, and efficient scale.

Companies with a narrow moat are those we believe are more likely than not to achieve normalized excess returns for at least the next 10 years. Wide-moat companies are those in which we have very high confidence that excess returns will remain for 10 years, with excess returns more likely than not to remain for at least 20 years. The longer a firm generates economic profits, the higher its intrinsic value. We believe low-quality, no-moat companies will see their normalized returns gravitate toward the firm's cost of capital more quickly than companies with moats.

When considering a company's moat, we also assess whether there is a substantial threat of value destruction, stemming from risks related to ESG, industry disruption, financial health, or other idiosyncratic issues. In this context, a risk is considered potentially value destructive if its occurrence would eliminate a firm's economic profit on a cumulative or midcycle basis. If we deem the probability of occurrence sufficiently high, we would not characterize the company as possessing an economic moat.

2. Estimated Fair Value

Combining our analysts' financial forecasts with the firm's economic moat helps us assess how long returns on invested capital are likely to exceed the firm's cost of capital. Returns of firms with a wide economic moat rating are assumed to fade to the perpetuity period over a longer period of time than the returns of narrow-moat firms, and both will fade slower than no-moat firms, increasing our estimate of their intrinsic value.

Our model is divided into three distinct stages:

Stage I: Explicit Forecast

In this stage, which can last five to 10 years, analysts make full financial statement forecasts, including items such as revenue, profit margins, tax rates, changes in workingcapital accounts, and capital spending. Based on these projections, we calculate earnings before interest, after taxes (EBI) and the net new investment (NNI) to derive our annual free cash flow forecast.

Stage II: Fade

The second stage of our model is the period it will take the company's return on new invested capital—the return on capital of the next dollar invested ("RONIC")—to decline (or rise) to its cost of capital. During the Stage II period, we use a formula to approximate cash flows in lieu of explicitly modeling the income statement, balance sheet, and cash flow statement as we do in Stage I. The length of the second stage depends on the strength of the company's economic moat. We forecast this period to last anywhere from one year (for companies with no economic moat) to 10–15 years or more (for wide-moat companies). During this period, cash flows are forecast using four assumptions: an average growth rate for EBI over the period, a normalized investment rate, average return on new invested capital (RONIC), and the number of years until perpetuity, when excess returns cease. The investment rate and return on new invested capital decline until a perpetuity value is calculated. In the case of firms that do not earn their cost of capital, we assume marginal ROICs rise to the firm's cost of capital (usually attributable to less reinvestment), and we may truncate the second stage.

Stage III: Perpetuity

Once a company's marginal ROIC hits its cost of capital, we calculate a continuing value, using a standard perpetuity formula. At perpetuity, we assume that any growth or decline or investment in the business neither creates nor destroys value and that any new investment provides a return in line with estimated WACC.

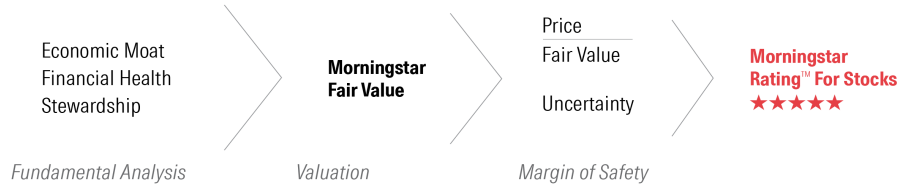
Because a dollar earned today is worth more than a dollar earned tomorrow, we discount our projections of cash flows in stages I, II, and III to arrive at a total present value of expected future cash flows. Because we are modeling free cash flow to the firm—representing cash available to provide a return to all capital providers—we discount future cash flows using the WACC, which is a weighted average of the costs of equity, debt, and preferred stock (and any other funding sources), using expected future proportionate long-term, market-value weights.

3. Uncertainty Around That Fair Value Estimate

Morningstar's Uncertainty Rating is designed to capture the range of potential outcomes for a company's intrinsic value. This rating is used to assign the margin of safety required before investing, which in turn explicitly drives our stock star rating system. The Uncertainty Rating is aimed at identifying the confidence we should have in assigning a fair value estimate for a given stock.

Our Uncertainty Rating is meant to take into account anything that can increase the potential dispersion of future

Morningstar Equity Research Star Rating Methodology



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outcomes for the intrinsic value of a company, and anything that can affect our ability to accurately predict these outcomes. The rating begins with a suggested rating produced by a quantitative process based on the trailing 12-month standard deviation of daily stock returns. An analyst overlay is then applied, with analysts using the suggested rating, historical rating data, and their own knowledge of the company to inform them as they make the final Uncertainty Rating decision. Ultimately, the rating decision rests with the analyst. Analysts take into account many characteristics when making their final decision, including cyclical factors, operational and financial factors such as leverage, company-specific events, ESG risks, and anything else that might increase the potential dispersion of future outcomes and our ability to estimate those outcomes.

Our recommended margin of safety—the discount to fair value demanded before we'd recommend buying or selling the stock—widens as our uncertainty of the estimated value of the equity increases. The more uncertain we are about the potential dispersion of outcomes, the greater the discount we require relative to our estimate of the value of the firm before we would recommend the purchase of the shares. In addition, the Uncertainty Rating provides guidance in portfolio construction based on risk tolerance.

Our Uncertainty Ratings are: Low, Medium, High, Very High, and Extreme.

	Margin of Safety	
Qualitative Analysis Uncertainty Ratings	★★★★★Rating	★Rating
Low	20% Discount	25% Premium
Medium	30% Discount	35% Premium
High	40% Discount	55% Premium
Very High	50% Discount	75% Premium
Extreme	75% Discount	300% Premium

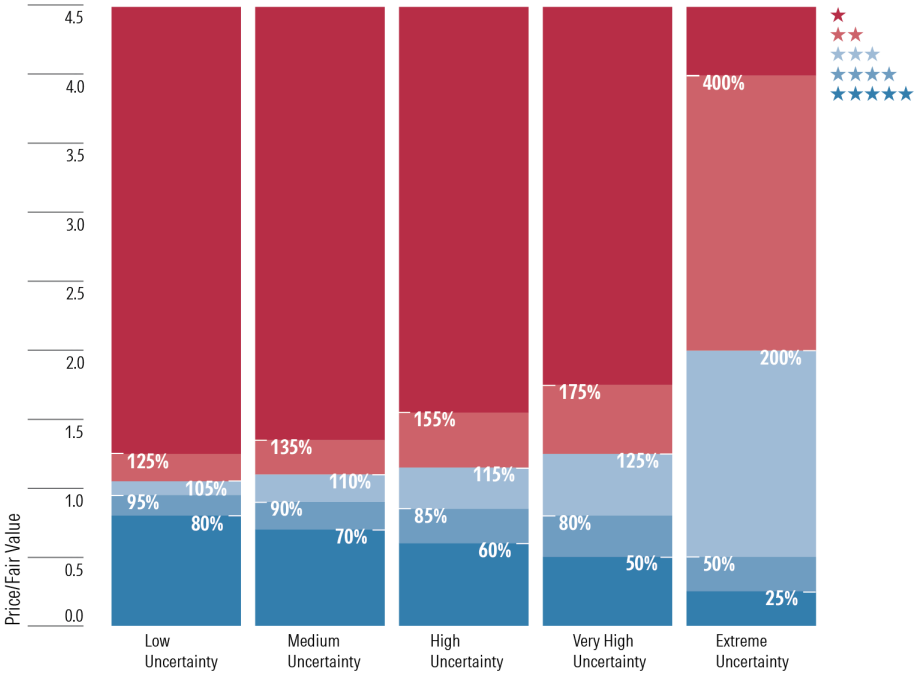
Our uncertainty rating is based on the interquartile range, or the middle 50% of potential outcomes, covering the 25th percentile–75th percentile. This means that when a stock hits 5 stars, we expect there is a 75% chance that the intrinsic value of that stock lies above the current market price. Similarly, when a stock hits 1 star, we expect there is a 75% chance that the intrinsic value of that stock lies below the current market price.

4. Market Price

The market prices used in this analysis and noted in the report come from exchange on which the stock is listed which we believe is a reliable source.

For more details about our methodology, please go to https://shareholders.morningstar.com

Morningstar Equity Research Star Rating Methodology



Morningstar Star Rating for Stocks

Once we determine the fair value estimate of a stock, we compare it with the stock's current market price on a daily basis, and the star rating is automatically re-calculated at the market close on every day the market on which the stock is listed is open. Our analysts keep close tabs on the companies they follow, and, based on thorough and ongoing analysis, raise or lower their fair value estimates as warranted.

Please note, there is no predefined distribution of stars. That is, the percentage of stocks that earn 5 stars can fluctuate daily, so the star ratings, in the aggregate, can serve as a gauge of the broader market's valuation. When there are many 5-star stocks, the stock market as a whole is more undervalued, in our opinion, than when very few companies garner our highest rating.

We expect that if our base-case assumptions are true the market price will converge on our fair value estimate over time generally within three years (although it is impossible to predict the exact time frame in which market prices may adjust).

Our star ratings are guideposts to a broad audience and individuals must consider their own specific investment goals, risk tolerance, tax situation, time horizon, income needs, and complete investment portfolio, among other factors.

The Morningstar Star Ratings for stocks are defined below:

★★★★★ We believe appreciation beyond a fair risk adjusted return is highly likely over a multiyear time frame. Scenario analysis developed by our analysts indicates that the current market price represents an excessively pessimistic outlook, limiting downside risk and maximizing upside potential.

★★★★ We believe appreciation beyond a fair risk-adjusted return is likely.

★★★ Indicates our belief that investors are likely to receive a fair risk-adjusted return (approximately cost of equity).

★★ We believe investors are likely to receive a less than fair risk-adjusted return.

★ Indicates a high probability of undesirable risk-adjusted returns from the current market price over a multiyear time frame, based on our analysis. Scenario analysis by our analysts indicates that the market is pricing in an excessively optimistic outlook, limiting upside potential and leaving the investor exposed to Capital loss.

Other Definitions

Last Price: Price of the stock as of the close of the market of the last trading day before date of the report.

Capital Allocation Rating: Our Capital Allocation (or Stewardship) Rating represents our assessment of the quality of management's capital allocation, with particular emphasis on the firm's balance sheet, investments,

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and shareholder distributions. Analysts consider companies' investment strategy and valuation, balance sheet management, and dividend and share buyback policies. Corporate governance factors are only considered if they are likely to materially impact shareholder value, though either the balance sheet, investment, or shareholder distributions. Analysts assign one of three ratings: "Exemplary", "Standard", or "Poor". Analysts judge Capital Allocation from an equity holder's perspective. Ratings are determined on a forward looking and absolute basis. The Standard rating is most common as most managers will exhibit neither exceptionally strong nor poor capital allocation.

Capital Allocation (or Stewardship) analysis published prior to Dec. 9, 2020, was determined using a different process. Beyond investment strategy, financial leverage, and dividend and share buyback policies, analysts also considered execution, compensation, related party transactions, and accounting practices in the rating.

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